

RESEARCH ARTICLE

Forecasting Floods Using Deep Learning Models: A Longitudinal Case Study of Chenab River, Pakistan

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ABSTRACT This study presents an integrated methodological approach for forecasting future floods. Over the course of human history, floods have undoubtedly been the most common form of catastrophe. In this study, hydrological analysis was combined with time-series modeling, and this integrated approach was used to forecast daily river flow. The daily runoff of the catchment area was calculated using curve number modeling by incorporating static spatial statistics and temporal rainfall data. The applicability of time-series modeling is checked through two deep learning algorithms: Long Short-Term Memory (LSTM) and Multilayer Group Method of Data Handling (ML-GMDH). This integrated approach yielded an optimal set of attributes for river flow prediction. The proposed methodology was applied to the Chenab River Basin, Pakistan, where the absence of machine learning and deep learning models tailored for flood prediction emphasizes the urgent need for innovative and advanced tools that facilitate timely flood forecasts. The LSTM model obtained the best results, attaining the highest R^2 of 0.91, and the highest coefficient of correlation of 0.96. Additionally, the ML-GMDH model signifies its capability with 0.88 R^2 . The findings of this study demonstrate that the proposed hybrid approach is a promising and reliable choice for river flow prediction, particularly for ungauged basins where upstream hydrological data is not available. The proposed model is advanced and effective, and can be employed to forecast future flood events in the Chenab Basin and similar locations worldwide.

INDEX TERMS Deep learning, flood forecasting, geospatial analysis, GMDH, LSTM, remote sensing, time-series modeling.

I. INTRODUCTION

Of all the natural catastrophes, floods are the most recurrent (46%) and cause the most human suffering and loss. The frequency of floods is double the rate at which other natural disasters occur, and their impact is about three times that of typical cyclones with regard to affecting the human population [1]. Floods cover more geographical areas than other disasters worldwide. The indirect losses are greater than the direct damage. Climate is changing worldwide owing to a number of factors, with global warming being the prime contributor. Climate change can pose a severe threat to mankind in the form of intense weather conditions and widespread

flooding [2]. In 2022, floods hit the world 176 times, which is more than any other calamity [3].

For flood forecasting, natural attributes can be effectively used to develop mathematical models in which the mutual relationship between attributes is represented in the form of an equation. Some examples of relevant natural attributes that act as input features for flood forecasting include rainfall, temperature, river discharge, land use, topography, and soil details. It is essential to understand the meteorological scenarios of an area to know the reasons behind flooding. Using climate variables in flood forecast modeling is a common practice among researchers [4]. According to Breugem et al. [5] flood events can be caused by meteorological conditions and result in hydrological penalties. Among all meteorological factors, rainfall is considered the prime factor for flooding

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and is a key input parameter for flood risk forecasting [6]. Severe rainfall over a short duration or a continuous series of rainfall events over a longer period can cause flooding [5]. In many studies [7], [8], [9], [10], [11], [12], the authors used numerous meteorological parameters for flood event simulation, while in some other studies [13], [14], [15], [16] rainfall was used as a single predictor variable for river flow prediction. Different types of physical catchment characteristics, land use, and soil respond differently to rainfall events. Grass, crops, and forests can allow more water to pass into the soil, whereas concrete surfaces do not [17]. Forested catchments are capable of lessening the effects of rainstorms, whereas barren lands are less capable of doing so [18]. Moreover, soil texture types are divided based on their potential to infiltrate water [17], [19], [20]. Meteorological parameters and physical catchment characteristics such as rainfall, temperature, elevation, land use, and soil texture details help simulate hydrological processes like evapotranspiration, infiltration, interception, sub-surface runoff, and snow melt, which in turn impact the process of flooding [21]. Flood forecasting methods may be either data-driven or physically based hydrological models, where the former type of model is generally less complex. In contrast, hydrological models require greater detail and deeper understanding, which results in a higher number of parameters in the underlying equation representing the relationship [22]. Data-driven studies primarily rely on rainfall and meteorological indices for flood forecasting. Statistical modeling methods often judge river flow based on hydrological data that may be upstream station flow or temporal flow of the same station [23], [24], [25]. Physically based hydrological models rely on mathematical equations to measure the physical properties of catchments and simulate the physical processes involved in flooding [22]. Table 1 lists some of the hydrometeorological and physical attributes used in different types of flood forecasting methods. A physically based hydrological model cannot capture the complexities and dynamics of flood nature [19]. Deep learning methods are powerful tools for modeling nonlinear and complex systems. Flood modeling often requires the handling of large datasets for reliable forecasting. Machine learning and deep learning approaches can effectively handle large sets of data [26]. They can be a worthwhile alternative to physically based models for forecasting floods. Recent studies have shown that artificial intelligence (AI) methods are more promising, cost-effective, and reliable choices for river flow prediction than conceptual models [27].

In this study, hydrological analysis is coupled with intelligent time-series modeling to form a new integrated approach aimed at forecasting future floods. Through this integration, an optimal set of attributes is generated. The study employs the integration of two deep learning algorithms “Long Short-Term Memory (LSTM),” and “Multilayer Group Method of Data Handling (ML-GMDH),” with traditional hydrological modeling. The traditional and popular “Natural Resource Conservation Service Curve Number (NRCS-CN) method”

is used for surface runoff modeling [19], [28]. The proposed hybrid method is applied to predict the daily river flow at the Marala Station, situated along the Chenab River in Pakistan. The Chenab River is transboundary where timely access to upstream hydrometeorological data is not available [7], [29]. It is difficult to predict floods in the absence of ground truth data. Pakistan’s government does not have a reliable information collection mechanism and an effective and timely flood forecasting system [30], [31]. Other than government department efforts, multiple researchers have attempted to overcome the challenge of effective flood forecasting in Chenab Basin, Pakistan. These studies are found to be based on hydrological modeling [7], [8], [29]. To the best of our knowledge, no machine learning or deep learning-based flood forecasting systems have been developed for Chenab Basin, Pakistan. Our proposed approach is advanced and effective, and employs deep learning algorithms for the study area for the first time. The results obtained with both models show their powerful capability to model nonlinear and complex systems. The overall findings of this study demonstrate the usefulness of the proposed approach. This study takes advantage of advanced and modern “geographical information system (GIS)” technology. In the absence of ground station gauge data, satellite-based products are a great advantage. Important data attributes for this study are imported from “Google Earth Engine (GEE)” [32].

The key contributions of this study are:

- Timely flood prediction through a novel approach of integrating hydrological rainfall-runoff modeling with data-driven time-series analysis.
- The study’s findings demonstrate the capability of both LSTM and ML-GMDH models to yield satisfactory results, underscoring their potential effectiveness in enhancing flood prediction efforts.
- The results signify that the proposed hybrid approach is a suitable choice for ungauged river basins and can effectively predict river flow using satellite-based meteorological and physical information in the absence of upstream hydrological data.
- Development of an advanced and efficient flood forecasting system employing deep learning algorithms for the Chenab River Basin, Pakistan, for the first time.
- Development of a cost-effective solution to the problem by using Python IDE and freely available satellite data.

The remainder sections of this paper have the following details. Section II covers some of the hydrological and data-driven efforts made for flood prediction in previous research. Section III covers the details of the study area, the datasets used in this research, their sources, and a step-by-step description of the proposed methodology. Section IV discusses the achievements, limitations, and future implications of the proposed study. Section V concludes the study’s methodology and findings.

TABLE 1. A comparison of flood forecasting methods with choice of attributes.

Ref	Framework	Model Type	Parameters used	Best Results
[7]	Physically-based hydrological modeling	Hydrological Engineering Centre-Hydrological Modeling System (HEC-HMS)	Satellite precipitation, land surface temperature, snow melt, land use, and soil details	$R^2 = 0.94$
[8]	Physically-based hydrological modeling	Integrated Flood Analysis System (IFAS)	Satellite precipitation, Land use, soil, topographic characteristics, and forecasted rain data	$R^2 = 0.84$
[33]	Physically-based hydrological modeling	Soil and Water Assessment Tool (SWAT)	Rainfall, temperature, humidity, wind speed, solar radiation, topography, land use, and soil texture details	$R^2 = 0.75$
[23]	Statistical modeling	ARIMA	River flow	$R^2 = 0.84$
[24]	Statistical modeling	ARIMA	River water level	Satisfactory
[13]	AI methods	Support Vector Machine, K Nearest Neighbor, Random Forest, Decision Tree	Rainfall	Accuracy = 85%
[14]	AI methods	ANN	Rain, river flow	$R^2 = 0.72$
[34]	AI methods	LSTM	20 variables including water level of river, rain, humidity, evaporation, water content in reservoirs, runoff, thermal radiation, solar radiation, atmospheric pressure, river discharge, temperature, last day runoff	$R^2 = 0.67$

II. LITERATURE REVIEW

Researchers have extensively used rainfall-runoff models for flood forecasting [19], [28]. Rainfall-runoff models are a common example of hydrological models used for flood forecasting. They focused on the generation of runoff from rainfall events. The traditional method of calculating rainfall-runoff is an event-based, lumped model based on empirical equations. The objective of these studies was to estimate the runoff depth of watersheds using the “Natural Resource Conservation Service Curve Number (NRCS-CN)” method [28], [35]. The popular distributed hydrological model HEC-HMS is widely used for flood forecasting in research with numerous meteorological and hydrological parameters and physical catchment characteristics, such as rain, river discharge, river water level, digital elevation model, land use details, soil type, and forecasted rain [7], [9]. The “Integrated Flood Analysis System (IFAS)” is a hydrological model based on the kinematic wave equation that has been developed as a tool for flood forecasting, especially in developing countries [8]. In these studies [29], [36], [37], the IFAS was regionally parameterized with satellite precipitation products, river discharge, and other geographical catchment attributes to obtain a forecast solution. The semi-distributed physically based hydrological model “Soil

and Water Assessment Tool (SWAT)” employs physical characteristics such as land use, soil texture, topographic details, and rainfall for flood monitoring [33], [38].

Data-driven approaches are simpler and becoming popular because of their ability to handle detailed sets of information related to various scientific fields [23], [24], [26]. It has been established that AI-based methods perform better for flood prediction [34]. Machine learning, deep learning, and reinforcement learning techniques are playing significant roles in many scientific fields like disease recognition [39], [40], game theory [41], and time-series forecasting [27], [34], [42]. Time-series modeling using machine learning and deep learning models is more cost-effective and better in terms of performance [16]. Lawal et al. [13] demonstrated the superiority of the regression algorithm over other machine learning algorithms for flood prediction with monthly rainfall data. Liu et al. [43] assumed that the relationship between rainfall and runoff is nonlinear, which cannot be well described by a regression model and used a neural network to accurately estimate the runoff. Mistry and Parekh [14] developed a single-layer feed-forward ANN and successfully applied it for river flow prediction. The LSTM model is a state-of-the-art benchmark method for time-series modeling. It is widely used in flood forecasting applications owing to its ability

to remember long-term dependencies [44]. The ability of LSTM to remember long-term dependencies provides it an edge over ANN and other machine learning methods. In [44], the authors used basic recurrent neural network (RNN) and LSTM models for river water quality prediction based on temporal water quality data. LSTM outperformed the RNN with a high-precision output. Atashi et al. [45] compared the performance of a statistical model “SARIMA,” a machine learning model “Random Forest” and LSTM to predict river water levels in the Red River of North. The LSTM showed more promising results than the other two approaches. Google’s operational flood forecasting system [27] has used hourly rainfall and upstream station water levels to predict downstream riverine floods since 2021 in widespread areas of India and Bangladesh. They opted for two methods: a linear model and a Long Short-Term Memory. Again, LSTM exhibited a higher performance. Faruq et al. [25] used the LSTM model in time-series modeling to forecast floods using a river water level dataset. The results proved that LSTM model is a promising choice for time-series modeling. Kratzert et al. [11] showed the power of a single LSTM model to be trained over 531 basins in the United States for runoff prediction and still outperformed the regionally trained conceptual models in those basins. Again, Kratzert et al. [46] used the LSTM model to successfully simulate runoff behavior over 241 catchments. The GMDH is another powerful identification technique that can be used to model unknown and complex relationships between variables in a system without having particular knowledge of processes [47]. It is a nonlinear, heuristic, and self-organizing neural network architecture implemented using a feed-forward neural network with multilayered bi-input neurons [48]. The authors in [49] proved the prediction ability of GMDH modeling in the case of non-stationary data. The GMDH forecasted the runoff of the Ghale-Chay River with high accuracy. Ebtehaj et al. [50] predicted the river water level based on the “Generalized Structure-Group Method of Data Handling (GS-GMDH)” and “Adaptive Neuro-Fuzzy Inference System with Fuzzy C-Means (ANFIS-FCM)”. The results showed that GS-GMDH predicted with better accuracy. GMDH was successfully employed by [47], [48], [51], and [52] in forecasting analysis through time-series modeling. In this study, we propose the integration of runoff modeling with time-series analysis for the purpose of flood monitoring.

III. MATERIALS AND METHODS

A. STUDY AREA-CHENAB RIVER BASIN

The Chenab River is one of the largest rivers in the subcontinent. The river, also known as the “Chandra Bhaga River”, is formed by the confluence of two streams: “Chandra” and “Bhaga.” Both streams originate in the upper Himalayas of India [53]. Chenab River is transboundary with 98% of its catchment area in India. The catchment range is mostly mountainous with steep slopes and high altitudes. Marala

is the first river water discharge measuring station on the Chenab River in Pakistan. The Chenab River catchment upstream of the Marala Barrage contributes flows to the Marala Barrage outlet, whose meteorological and geographical characteristics are analyzed in this study. Until the Marala Barrage, the river covered a total distance of 506 km [54]. The total length of the river is 274 km downstream of Marala with a basin area of 41,656 km². The total catchment area upstream of Marala is approximately 28,480 km² and consists of 20 sub-basins [55]. The geographical catchment area of the Chenab River lies between 74°-78° E and 32°-35° N. The elevation of the catchment area ranges between 217 m and 7,101 m as shown in Fig. 2. The slope varies from 25 meters per kilometer in the upper Himalayan part to a 0.4 meters per kilometer average slope downstream of Marala in flood plains. The river has an average annual discharge of 1.52 million m³ [56]. The floods in Chenab are a result of massive precipitation in the upstream catchment area, which lies in the hub of the monsoon belt between the Arabian Sea and Bay of Bengal. Around South Asia, this season extends from July to September. The Chenab River catchment is a snow-dominant area. The monsoon and pre-monsoon seasons contribute to approximately 65% of the precipitation and only 26% of the precipitation results in the form of snow during winter.

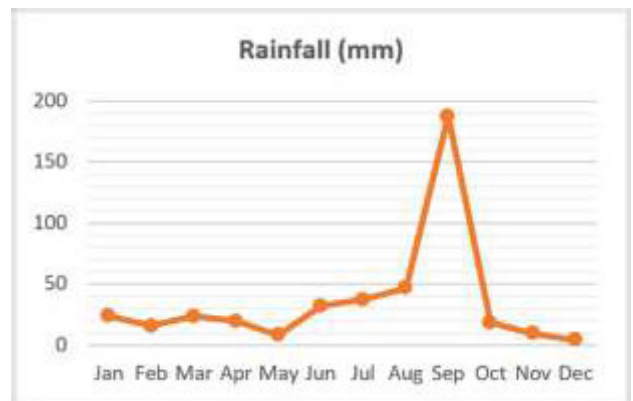


FIGURE 1. Monthly maximum Rainfall of river Chenab catchment for the devastating flood year 2014.

Therefore, the river receives its peak flow during the monsoon season, which is aggravated by overlapping snowmelt during the summer [7]. In addition, the higher flows from tributaries flowing into the river also add to the river water discharge. There is a constant variation in the temperature throughout the year. May and June are the hottest months, whereas December and January are the coldest ones. Fig. 1 shows the pattern of rainfall over the Chenab catchment area during the 2014 flood year.

Rainfalls of 187 and 141 mm were recorded on September 4 and 5, 2014, respectively, which caused a widespread and devastating flood in the Chenab River floodplains. According to “Pakistan Meteorological Department (PMD),” as a result

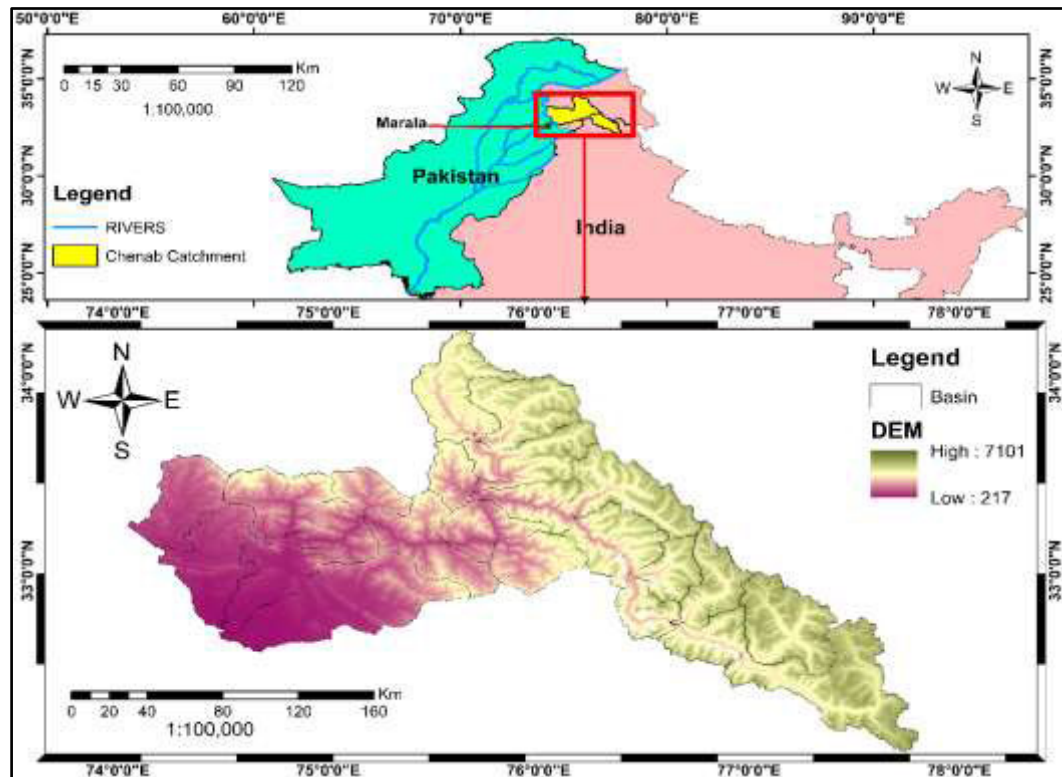


FIGURE 2. Visualization and elevation of the study area: 98% of Chenab catchment lies in India while Marala is Pakistan's first river discharge measuring station on Chenab River near the border of Pakistan. Due to High altitude, catchment upstream of the Marala Barrage contributes flows to the Marala Barrage outlet.

of this massive rainfall, floods crossed the 5,62,848 cusecs flow at Marala Station on September 7, 2014 [57].

B. DATASETS USED

Flood forecasting in the Chenab River Basin requires high-resolution and reliable hydrometeorological data, land use, and soil information [7]. This is always a challenge in transboundary river basins because of the insufficient data sharing between countries. The meteorological, hydrological, and physical attributes collected for this study were river flow, rainfall, temperature, surface pressure, humidity, soil texture, and land use details of the study area. All hydrometeorological attributes were collected for a period of 13 years from 2010 to 2022 on a daily time scale. Except river flow all other variables are imported from Google Earth Engine (GEE). It is a cloud-based platform with a large archive of freely available remote sensing and GIS datasets. Its interactive cloud computing platform allows users to develop different geospatial algorithms, manipulate and visualize existing datasets, map changes and trends on Earth's surface, and quantify their differences [32]. Multiple sources of data available on the GEE were examined for selected features. The methodology was then finalized using the dataset sources described below.

Land cover statistics are calculated using the dataset "ESA WorldCover 10m v100" Global Land Cover Map developed and validated in near real-time, based on Sentinel-1 and Sentinel-2 satellites data freely available on the GEE platform [58]. It is a product of the European Space Agency (ESA) with a spatial resolution of 10 m [59]. The dataset used for the identification of the hydrological soil group is "OpenLandMap Soil Texture Class (USDA System)" available on GEE's platform with a resolution of 250 m [60]. OpenLandMap services are provided by the OpenGeoHub Foundation to provide open access to global land resources and soil datasets [61]. The USDA System of Soil Classification was developed by the "United States Department of Agriculture (USDA)" [62], [63]. The rainfall dataset used for rainfall input and creation of antecedent moisture conditions is "CHIRPS Daily: Climate Hazards Group InfraRed Precipitation with Station Data (Version 2.0 Final)" at a spatial resolution of 0.25° [64]. For meteorological parameters, land surface temperature, atmospheric pressure, and humidity, "ERA5-Land Daily Aggregated - ECMWF Climate Reanalysis" dataset is explored [65]. Here, ECMWF stands for "European Center for Medium-Range Weather Forecasts." ERA5 is the product of the ECMWF's implemented program "Copernicus Climate Change Service (C3S)" [66]. The daily river flow of Marala hydrological station was

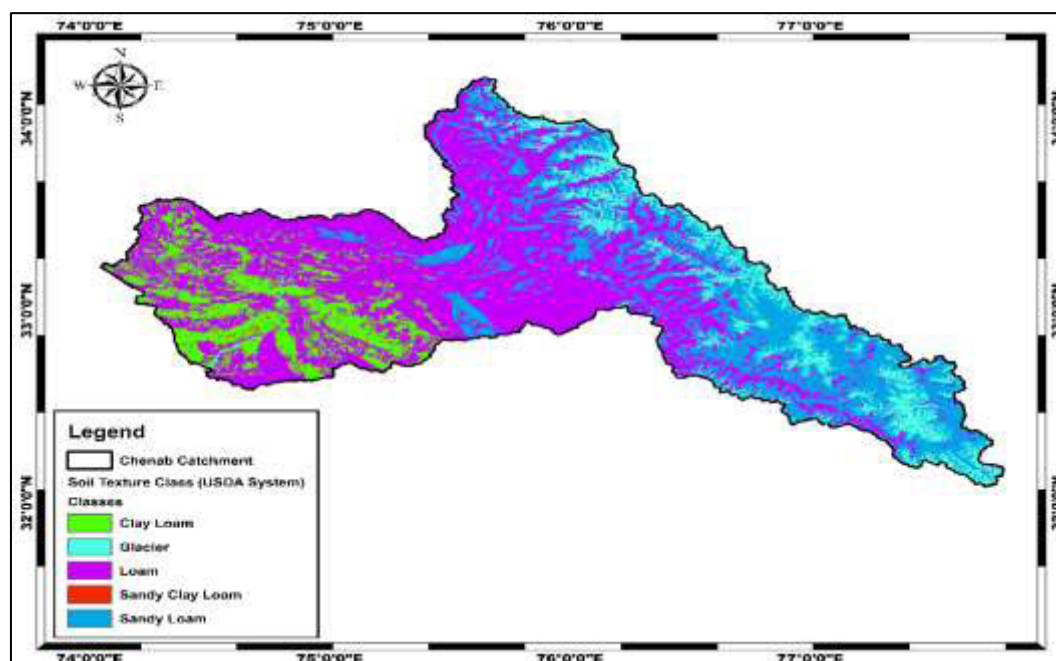


FIGURE 3. Digital identification of hydrological soil groups available in Chenab River catchment using “OpenLandMap Soil Texture Class (USDA System)” dataset. Four types of soil textures are found in the study area. Soils found in the study area have moderate water infiltration capacity.

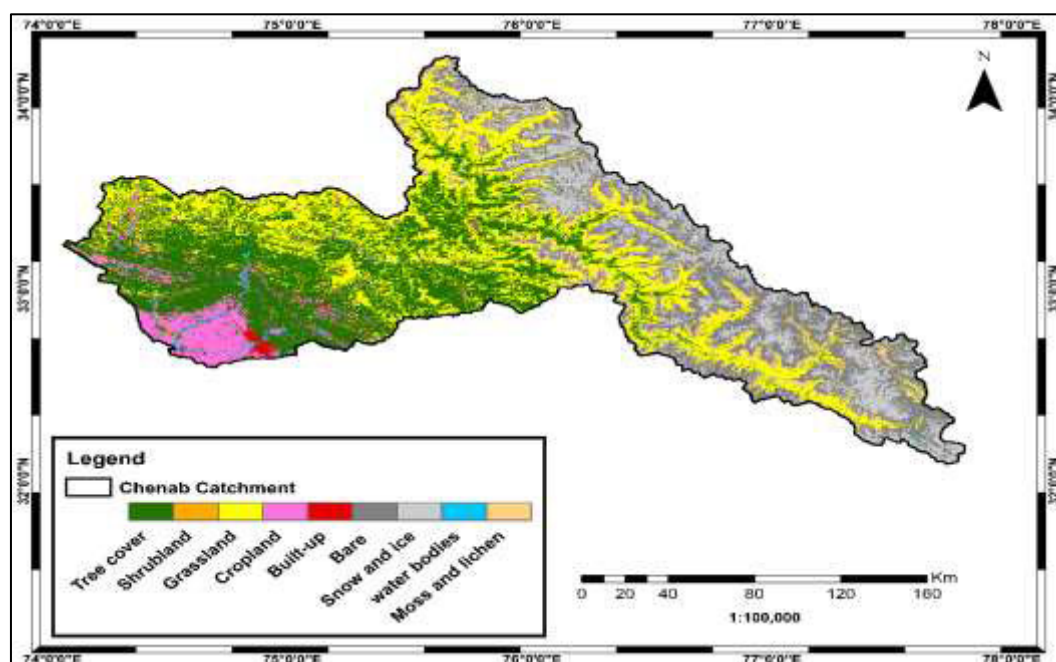


FIGURE 4. Land-use map of Chenab catchment using ESA's Global Land Cover Map at a spatial resolution of 250m. Barren land and snow cover a large part of the study area, both classes have high runoff potential and low infiltration capacity.

acquired from the “Flood Forecasting Division, Pakistan Meteorological Department (FFD-PMD).” A shape file was used to import all types of information from the GEE. The geometric preparation of the Chenab catchment shape file was performed in ArcGIS 10.2, with the help of a standard

river Chenab catchment area shape file provided by the flood forecasting department (FFD) of Pakistan in JPG format and world hydro shed shape files freely available on the Internet [67]. The Digital Elevation Model (DEM) of the catchment area shown in Fig. 2, is prepared with “Shuttle Radar

Topography Mission (SRTM) DEM data at a spatial resolution of 30 m using ArcMap application. SRTM is a reliable and widely used source of DEM data for almost any location worldwide [68], [69].

1) SOIL INFORMATION

Hydrological soil groups are vital factors for the calculation of runoff depth and soil infiltration potential. The USDA has categorized the “Hydrologic Soil Groups (HSGs)” into four categories, A, B, C, and D depending on their potential to infiltrate [19], [70]. The NRCS method can calculate soil infiltration capacity even after continuous and long rainfall events [20]. The basic characteristics of the hydrological soil groups are listed in Table 2. Soil type statistics were calculated using the GEE code editor with the “OpenLandMap Soil Texture Class (USDA System)” dataset [71]. This dataset comprises 12 soil types, of which four types of soil textures were found in the study area.

TABLE 2. Hydrological soil groups and their characteristics.

Soil Group	Characteristics
Group A	These soils have high infiltration margins and the lowest runoff potential as compared to other groups even when thoroughly wetted.
Group B	These soils have better runoff potential as compared to group A and moderate infiltration rates when thoroughly wetted.
Group C	These soils have moderately low infiltration rates when thoroughly wetted and moderately high runoff potential.
Group D	These soils have the lowest infiltration rates and their runoff potential is highest.

These were clay loam, sandy clay loam, loam, and sandy loam (Fig. 3). Sandy loam belongs to HSG-B, and clay loam, sandy clay loam, and loam belong to HSG-C. It is concluded that only two hydrological soil groups, B and C, were present in the catchment area. Groups B and C have moderate infiltration capacity. The soil map was then imported into the ArcMap application to prepare the map legend (Fig. 3).

2) LAND-USE LAND-COVER (LULC) DETAILS

The land-use land-cover (LULC) characteristics of a catchment area significantly affect the percentage of streamflow rise during storm events. Statistics of land use classes found in the study area are calculated with the “ESA WorldCover 10m v100” dataset [58]. ESA land use data have also been used in many other hydrological simulation studies [7], [72]. The dataset includes 11 land cover classes: trees, shrubs, grass, crops, built-up land, barren land, snow, water bodies, herbaceous wetlands, moss and lichen, and mangroves. Except for mangroves, all classes were present in the study area as shown in Table 4. “Herbaceous wetlands” class is

found to have only 1 pixel (Table 4) at 250m resolution, it could not be viewed through the LULC map shown in Fig. 4. Because of its very high altitude, a large part of the catchment area is covered with snow and ice. Barren land constitutes a major portion of the watershed. Barren land, water bodies, and ice glaciers have a high runoff potential and low infiltration capacity [35], [73]. A map showing the land use classification of the study area was prepared on the GEE platform at 250m spatial resolution. It was then imported into the ArcMap application to prepare an attractive map legend. In the next section, we incorporate image static spatial statistics into rainfall-runoff modeling for a comprehensive analysis.

C. BRIEF OVERVIEW OF METHODOLOGY

This research was conducted to find a competent and robust solution for foreseeing floods and was specifically applied in the Chenab basin, Pakistan. Our proposed methodology is a hybrid technique that combines hydrological and intelligent time-series modeling. The methodology is briefly described in the below-mentioned steps.

Step 1: Various meteorological, hydrological, and static physical catchment attributes were collected to build a comprehensive dataset for this study. Daily data of rainfall, temperature, humidity, surface pressure of the Chenab catchment, and river flow of the Marala hydrological station were collected for a period of 13 years, spanning from January 2010 to December 2022. A detailed description of each dataset is provided in the “Datasets Used” section.

Step 2: Hydrological Rainfall-runoff modeling is implemented by incorporating temporal data of rainfall, antecedent rainfall, and spatial statistics of land use and soil. The popular NRCS-CN method is used for runoff calculation. This is a simple, conceptual method based on empirical equations [21]. The main objective of rainfall-runoff modeling is to calculate the percentage of rainwater that becomes part of the surface runoff. The initial phase of the process provides a weighted curve number for the entire catchment, which is a coefficient that estimates the runoff percentage from total precipitation [17]. This hydrological phase of the methodology generated the daily runoff depth parameter of the study area for 13 years from January 2010 to December 2022. Runoff is that percentage of rainwater that runs across the land and moves to the rivers. It is the excess water that cannot infiltrate or evaporate during a storm event. In this study, we attempted to map the river flow using runoff variable instead of rain. The results of our technique demonstrated the capability of our model.

Step 3: Time-series modeling is performed using two deep learning methods LSTM and ML-GMDH. A comprehensive daily dataset for 13 years was prepared for this phase of modeling to map daily river flow. The runoff calculated in the previous CN modeling step was used as an input parameter to calibrate both models, whereas the other meteorological parameters used were the temperature, atmospheric pressure, and humidity of the catchment area. Deep learning models

were capable of learning the temporal dependencies of information within time-series data and use this information to make better decisions regarding future values.

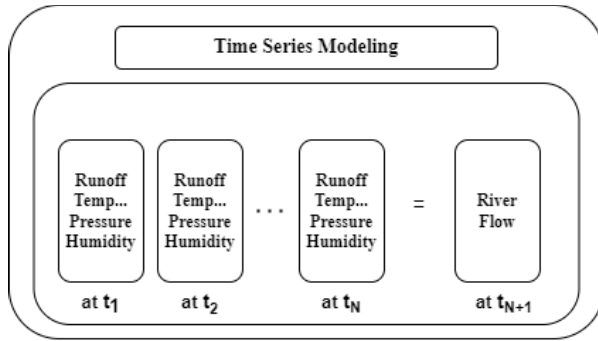


FIGURE 5. Time-series modeling with attributes used in the proposed research.

The main challenge in this study was to predict river flow data at a specific timestep $N + 1$ from the last N consecutive timesteps of the input sequence M .

Given an input sequence, $M = [X[1], X[2], \dots, X[N]]$ with N timesteps, where each element $X[N]$ is a vector containing input features (runoff, temperature, atmospheric pressure, and humidity) at the timestep t ($1 \leq t \leq N$). The last N consecutive timesteps are called look-back window.

In mathematical form, this relationship can be expressed as in equation (1).

$$R(t_{N+1}) = f(X(t_1), X(t_2), \dots, X(t_N)) \quad (1)$$

where future river flow (R) at time t_{N+1} is a function of the input feature vector (X) at times $t_1, t_2, \dots, t_N$.

Fig. 5 provides a glimpse of the time-series modeling with the attributes used in the proposed research. A detailed description of hydrological modeling is given in Section III-D, and time-series modeling is given in Section III-E. The methodological architecture used in this study is illustrated in Fig. 6

D. SPATIAL MODELING OF SURFACE RUNOFF

In this section, the traditional and popular “Natural Resource Conservation Service Curve Number (NRCS-CN) method” is used for surface runoff modeling. NRCS-CN runoff modeling is a function of rainfall, antecedent rainfall, soil, and land use details of the catchment area. The main objective of runoff modeling is to calculate the percentage of water that becomes a part of the surface runoff after rainfall. The curve number is a coefficient used in empirical equations to estimate the runoff potential from total precipitation [17]. The higher CN value depicts the higher runoff potential.

Each unique group of soil and land use found in any specific area responds differently to rainstorms and thus has a specific curve number assigned to it. An Intersection of the soil texture map and land use map was generated at 250m resolution to obtain joint statistics of soil and land use of the catchment area using the GEE platform. These joint statistics

were employed in the runoff modeling. An appropriate curve number value was assigned to each unique combination of available soil and land use in the catchment area. The curve numbers used in this study were obtained from the literature published by the USDA and other researchers [35], [70], [73], [74], [75], [76]. The weighted curve number was calculated based on the statistics shown in Table 4 using the NRCS-CN modeling method given in equation (2). Equation (2) is a weighted curve number formula used for the entire catchment area [28], [74]. Where, CN is the weighted Curve Number calculated for the entire catchment, A represents the total catchment area, i is the Index of catchment subdivisions of uniform soil and land use, A_i is the drainage area of subdivision i , and CN_i is the curve number of subdivision i .

$$CN = \frac{\sum (A_i \times CN_i)}{A} \quad (2)$$

The weighted curve number value shows that the Chenab catchment area has high runoff potential, which can be utilized for harvesting purposes to supplement irrigation Canals, and groundwater to increase agricultural productivity. The curve number, we have calculated above is termed as curve number for antecedent moisture condition AMC-II which is $CNII$. AMC represents the index of relative dryness or wetness of a catchment area prior to a storm event. It varies continuously and affects the runoff process considerably. The other two moisture conditions are dry and moist that is AMC-I and AMC-III respectively. The curve number is adjusted according to the soil moisture condition and wetness of watershed area by taking five days preceding rainfall data for each day. The criteria for AMC-I, AMC-II, and AMC-III are listed in Table 3 [77].

The values of the curve number suitable to the cases of AMC-I and AMC-III were calculated using equations (3) and (4) given by Hawkins et al. in 1985 [77].

$$CNI = \frac{CNII}{2.281 - 0.0128CNII} \quad (3)$$

$$CNIII = \frac{CNII}{0.427 - 0.00573CNII} \quad (4)$$

The CNI and $CNIII$ values are found to be 65.516 and 91.046, respectively. The NRCS-CN method followed the rainfall-runoff relationship presented in equation (5).

$$Q = \frac{(P - I_a)^2}{P - I_a + S} \quad (5)$$

where the runoff (Q) can be estimated by precipitation (P), potential maximum retention (S) and initial abstraction (I_a). The potential maximum retention (S) is the representation of infiltration which occurs after runoff has started. Potential maximum retention (S) can be calculated using equation (6). The Initial abstraction is defined in literature as $I_a = 0.2S$ [22], [28], [74]. It shows the water loss due to infiltration and interception before runoff starts.

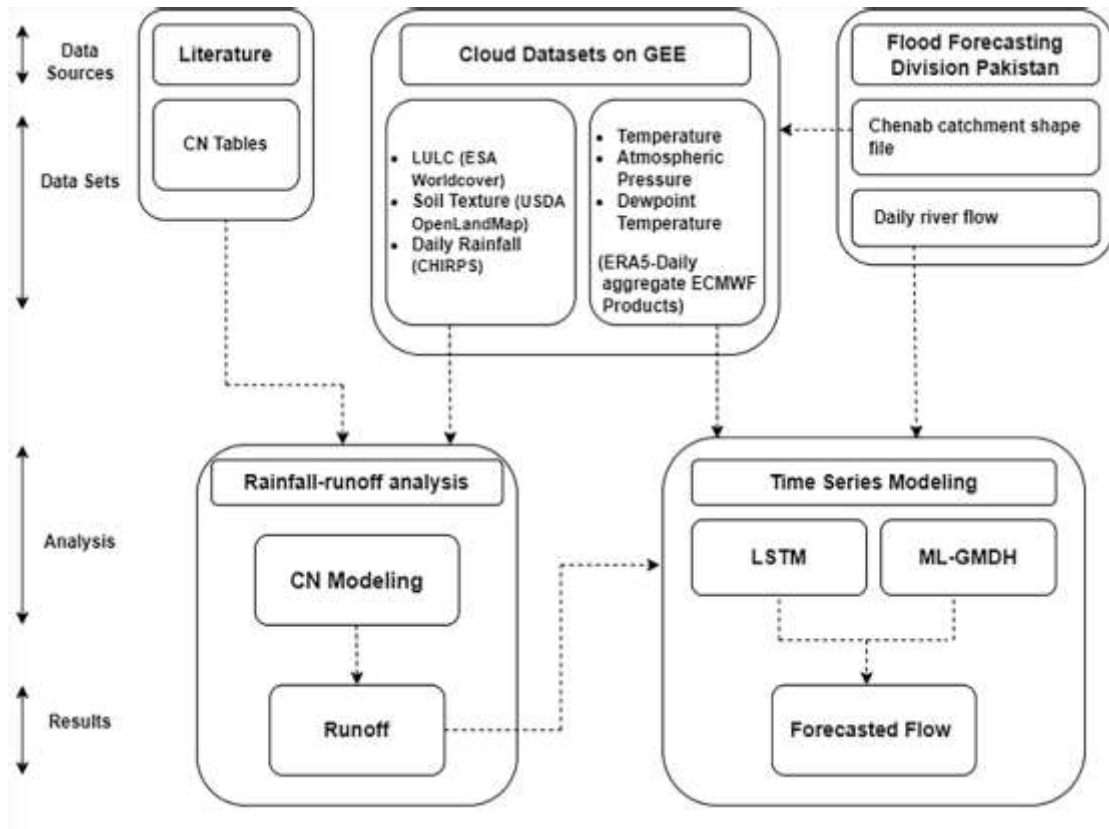


FIGURE 6. Proposed hybrid Methodology adopted for flood forecasting in this research.

TABLE 3. Criteria for antecedent moisture condition.

AMC	Total Rain in previous five days
I	Less than 12.7
II	Greater than and equal to 12.7 and less than 27.94
III	Greater than 27.94

The relationship between CN and S is presented in equation (6).

$$CN = \frac{25400}{254 + S} \quad (6)$$

CN for equation (6) can be CN_I or CN_{III} depending on daily AMC condition.

Daily rainfall data and statistics of land use and soil texture were imported in CSV format from Google Earth Engine for runoff analysis, and all calculations were performed in Python using the Pandas and NumPy libraries for this section. The integrated development environment used was the PyCharm Community Edition 2022.3.2.

This physical modeling has a significant impact on accurate river flow prediction using deep learning models. In the next section, we perform time-series analysis using two deep learning models to predict the river flow. The runoff calcu-

lated in this section was used as an input feature to train the time-series models.

E. DEEP LEARNING ANALYSIS

We performed time-series forecasting using two deep learning models, Long Short-Term Memory (LSTM) and the Multilayer Group Method of Data Handling (ML-GMDH), with the core objective of river flow or flood forecasting at the Head Marala station. In the data preprocessing phase, the following actions were performed to make the data valuable for decision-making, simplify computations, and increase the accuracy of the results. All variables were normalized using the Scikit-Learn library function MinMax Scaler. Normalization is essential to ensure that all the variables are at similar scales. Missing and incorrect values were identified and filled in using the mean imputation method. Python 3.10 programming language was used to create the deep learning algorithms. Scikit-Learn, NumPy, Pandas, and other popular Python machine-learning tools have been used to solve many programming-related queries, such as the arrangement and handling of data and solving machine learning difficulties. The Matplotlib library was used to plot the flow simulation and the train test loss graphs. The integrated development environment was the PyCharm Community Edition 2022.3.2.

The overall accuracy of the developed networks for the estimation of flood discharge was assessed using common indices such as the Coefficient of Determination (R^2),

TABLE 4. Land use and hydrological soil group joint statistics of the study area at 250m resolution.

LULC	HSG	No of pixels	Pixels (%)	Curve Number (CN)
Trees	B	16147	2.923	60
	C	143957	26.059	73
Grass	B	28943	5.239	69
	C	114424	20.713	79
Crops	B	25	0.005	78
	C	29773	5.389	85
Shrub and Scrub	B	0	0.000	72
	C	3312	0.600	81
Built	B	31	0.006	88
	C	2965	0.537	91
Barren	B	67024	12.132	86
	C	30768	5.570	91
Moss Lichen	B	14971	2.710	71
	C	16958	3.070	81
Herbaceous wetland	B	1	0.000	58
Snow and Ice		79218	14.340	100
Water		3919	0.709	100
Total		552436	100	81.28 (Weighted CN)

Pearson's r (R), Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). RMSE is the measurement of the square root of the average squared difference between the estimated and observed values for each data point of a dataset. A lower RMSE value indicates better performance of the model. The MAE is an important method for measuring the losses in regression-based algorithms. It calculates the mean of the absolute difference between actual and predicted values. R^2 calculates the amount of variation in the predicted variable that is foreseeable from the observed variable in regression-based models. It is a common metric used to assess the goodness of fit of regression models. Its value lies between zero and one, where a higher value indicates a smaller amount of variation between the predicted and observed values. Pearson's r (R) is used to calculate the strength and direction of the linear relationship between actual and predicted values. It ranges from -1 to 1 , where 1 reflects a perfect positive correlation between variables, -1 shows a perfect negative correlation, and 0 indicates that no linear correlation exists between the variables.

Equations (7), (8), (9), and (10) describe the aforementioned metrics.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (O_i - F_i)^2}{n}} \quad (7)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (O_i - \bar{O})^2}{\sum_{i=1}^n (O_i - \bar{O})^2} \quad (8)$$

$$R = \frac{\sum_{i=1}^n (O_i - \bar{O})(F_i - \bar{F})}{\sqrt{\sum_{i=1}^n (O_i - \bar{O})^2 \sum_{i=1}^n (F_i - \bar{F})^2}} \quad (9)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |F_i - O_i| \quad (10)$$

Among the metrics discussed above, O is the actual observed value, $\bar{O}$ is the mean of actual observed values, F is the predicted value, $\bar{F}$ is the mean of the predicted values and n represents the total number of values.

1) LONG SHORT-TERM MEMORY (LSTM)

LSTM is a renowned deep learning model. It is popular because of its ability to handle complex patterns in time-series data efficiently [34]. The LSTM architecture encompasses a feedback loop that can learn temporal dependencies between data and use the knowledge of past inputs in the analysis of current inputs. The LSTM architecture comprises special units called memory cells. Each LSTM cell is a logical unit. At each timestamp, the cell can decide what to do by using the state vector. There are three gates in the LSTM cell: the forget, input, and output gates. Information is added and removed to the cell state via gates that regulate information flow in the cell [78]. The purpose of the forget gate is to decide at each cell state which information coming from the previous timestamp is important enough to remember and which should be discarded. The input gate determines whether new information is important for storage and creates a new candidate value. Finally, the output gate filters unnecessary information and passes the final output value, which acts as the input for the next cell state [34]. The memory cell is a layer of neurons in a traditional feedforward neural network, where each neuron has a hidden layer and a current state. This architecture enables the LSTM to remember useful information over a longer period. The LSTM cell architecture and flow of information in the cell is illustrated in Fig. 7.

The model was designed based on the Keras framework, using the backend TensorFlow library. Experiments were conducted to select the most optimal input features. Various climatological satellite products were tested and some of them were found to be slightly correlated or not correlated with the target variable. The final selection was made by running multiple simulations of the LSTM model, where the coefficient of determination (R^2) was used as an objective function. The methodology was finalized with four parameters that are runoff, temperature, atmospheric pressure, and humidity to map the river flow. A complete dataset was prepared for further analysis that contained the daily values of the selected features for 13 years from 2010 to 2022.

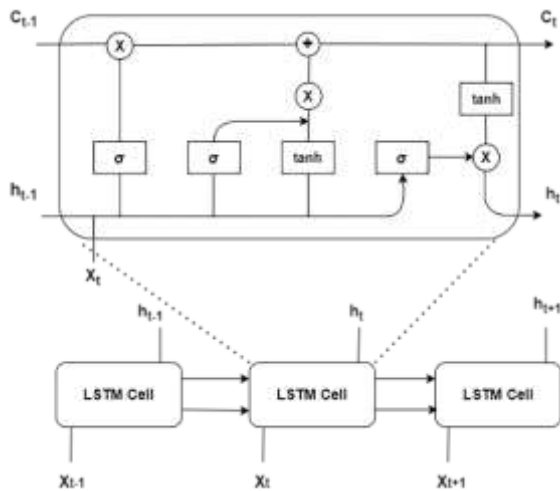


FIGURE 7. Flow of information in LSTM Cell Architecture.

This optimal train test ratio was identified again by running multiple simulations of the LSTM model. The dataset contained daily time-series data of 13 years, for which the training dataset contained data for 11 years, from 2010 to 2020 and the testing dataset has 2 years data, spanning over 2021 and 2022. Thus, 85% of the dataset was used for training, and 15% was used for testing purpose. The effects of simulations with 70%, 80%, and 90% of the training dataset were compared with the optimal selected ratio. A train test ratio of 85:15 performed slightly better.

Correct selection of the optimizer can increase the training speed and performance of the model. The performance of a machine learning model depends on its learning rate and weight. During the learning phase, methods known as optimizers are used to update the values of the learning rate and weights in each iteration. The combination of the loss function and optimizer used in the model is of high importance [16]. The loss function evaluates and guides the optimization performance by estimating the difference between the real and predicted values. We have compiled the LSTM model with the “Mean Absolute Error” Loss function and two optimizers, which are “Adaptive Moment Estimation (Adam),” and “Stochastic Gradient Descent (SGD),” where the Adam optimizer predicts with minimum loss. Fig. 8a-b shows the LSTM model performance with the MAE Loss function and both optimizer methods.

The number of previous timesteps that can best predict the next successive timestep is important for obtaining an optimized prediction. In the context of a hydrometeorological scenario, severe rainfall over a short duration and a continuous series of rainfall events spread over many days can cause extreme river flow events. The model was trained with various look-back window sizes (from 1 to 10 days) to identify the optimized window size.

The results of the statistical metrics indicated that meteorological observations over the last four days had the most

TABLE 5. Optimal parameters used in Final LSTM model.

Parameters Used	Values
Optimizer	Adam
Error functions	MAE and RMSE
Evaluation metrics	R^2 and Pearson's R
Batch size	16 - 128
Epochs	250 - 350
LSTM layers	1
LSTM Units	100
Dense layers	2
Dense Units	30 and 1

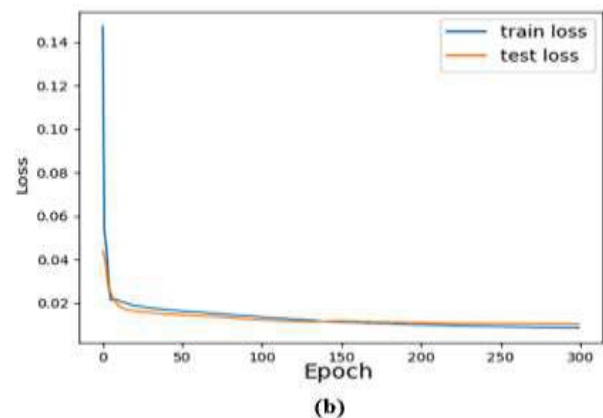
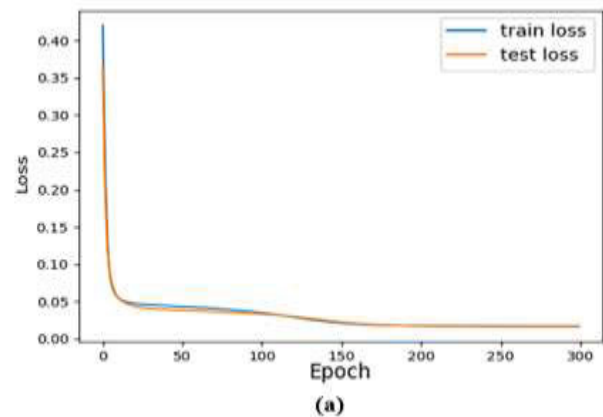


FIGURE 8. (a) LSTM model loss (MAE) with SGD optimizer (b) LSTM model loss (MAE) with Adam optimizer.

significant relationship with river flow on the fifth day. In Table 6, the results show that look-back periods of two, three, five, and six days are unable to satisfactorily explain the relationship between the input features and river flow.

The better performance of a deep learning model depends on hyperparameter selection. Network width and depth play a crucial role in the prediction accuracy. The final selected

architecture of the LSTM model after the test run was (100, 30, 1). That is, one LSTM layer with 100 units, one fully connected dense layer with 30 units, and Relu activation, and one fully connected dense layer with one unit and Sigmoid activation. One unit in the last dense layer represented the river discharge value for the next day as output. The network revealed that more than 300 epochs (350 in some cases) showed overfitting; here, the loss gradually increased with increasing number of epochs. More than 30 dense layers apparently had no positive effect on the model performance. Batch sizes of 16, 32, 64, 72, 100, and 128 successfully explain the relationship between the predicted and observed variables. Table 5 presents the parameters selected for the final LSTM model.

2) MULTILAYER GROUP METHOD OF DATA HANDLING (ML-GMDH)

The multilayer group method of data-handling (ML-GMDH) neural network is a multilayer iterative neural network. It is an unsupervised machine learning approach that is capable of reducing the dimensionality of data. The model creates polynomial equations or mathematical models that build a heuristic relationship between the input and output variables [51]. The model has the capability to select its structural parameters itself, such as the best number of input layers, the optimized number of neurons in the hidden layers, and important input variables [48]. During the training process, the network continuously self-organizes its structure by discarding bad neurons and selecting useful neurons in each layer [52]. Owing to this continuous process of removing bad neurons in each layer, the prediction results improved. In the ML-GMDH, a multilayer network of second-order polynomials (quadratic neurons) describes the complicated nonlinear relationships between the input and output features of a system [51]. In a quadratic neuron, a quadratic function is used to replace the inner product of the input feature vector and its respective weights in a traditional neuron [79].

During the training process, all possible combinations of input pairs are created using the following formula:

$$\frac{V!}{(V-2)!2!} \quad (11)$$

where V = number of input variables.

Each quadratic neuron in the GMDH network has two inputs and generates a single output. The output of a quadratic neuron can be estimated using equation (12) [51].

$$G = w_0 + w_1p_1 + w_2p_2 + w_3p_1p_2 + w_4p_1^2 + w_5p_2^2 \quad (12)$$

where p_1 and p_2 are two input variables, and w_i are the weights to be learned by quadratic neuron G and $w_i : i = 0, \dots, 5$.

As a result of this training process, a set of weights is produced for each neuron. The neurons with the smallest test errors are selected for all variables. These are sent to the next layer as input variables. This procedure is repeated for the next layer. When the errors in the test data in each layer

stop decreasing, the iterative calculation is terminated, that is how the multilayered architecture is organized [51]. The ML-GMDH architecture is illustrated in Fig. 9. By repeating this inherent, survival competition, and evolutionary process, the process continues until the newly produced neurons perform better than the previous ones and the heuristic model is selected.

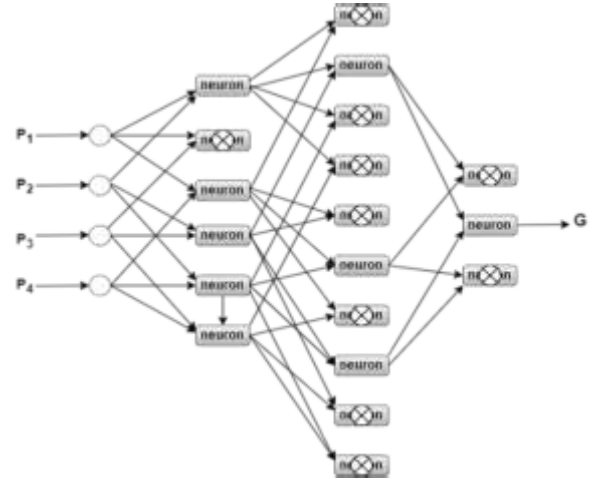


FIGURE 9. ML-GMDH model architecture explains with 4 inputs and 3 layers model.

In the second experiment, river discharge at Marala station was predicted using the ML-GMDH neural network. The model had four inputs and one single output. The optimal set of input features selected through LSTM network simulation experiments was used to map the daily river flow in this second experiment. The input features were daily runoff, temperature, atmospheric pressure, and humidity. The percentage of observations used for training and testing was the same as that in the case of the LSTM model, that is, 85% of the data were used for training and 15% for testing. The quest for an optimal architecture and optimal hyperparameter selection is not a struggle in the case of ML-GMDH modeling, as the model has the capability to select the optimized number of input layers, the best number of neurons in each layer, and important input variables itself.

IV. RESULTS AND DISCUSSIONS

In this section, the performance evaluation of both time-series models is discussed. The LSTM model achieves noteworthy results. The performance assessment of LSTM model with multiple parameter combinations and various look-back window sizes is presented in Table 6. The table clearly shows that the most promising results are achieved when the look-back window size is four.

The predicted river flow in trials 3, 4, 5, and 6 showed reasonable agreement with the observed flow, with higher values of R^2 and R , and lower RMSE. Trial 5 showed the highest values of R^2 and R that is 0.91 and 0.96, respectively, and the lowest RMSE of 9870, which is the best result achieved

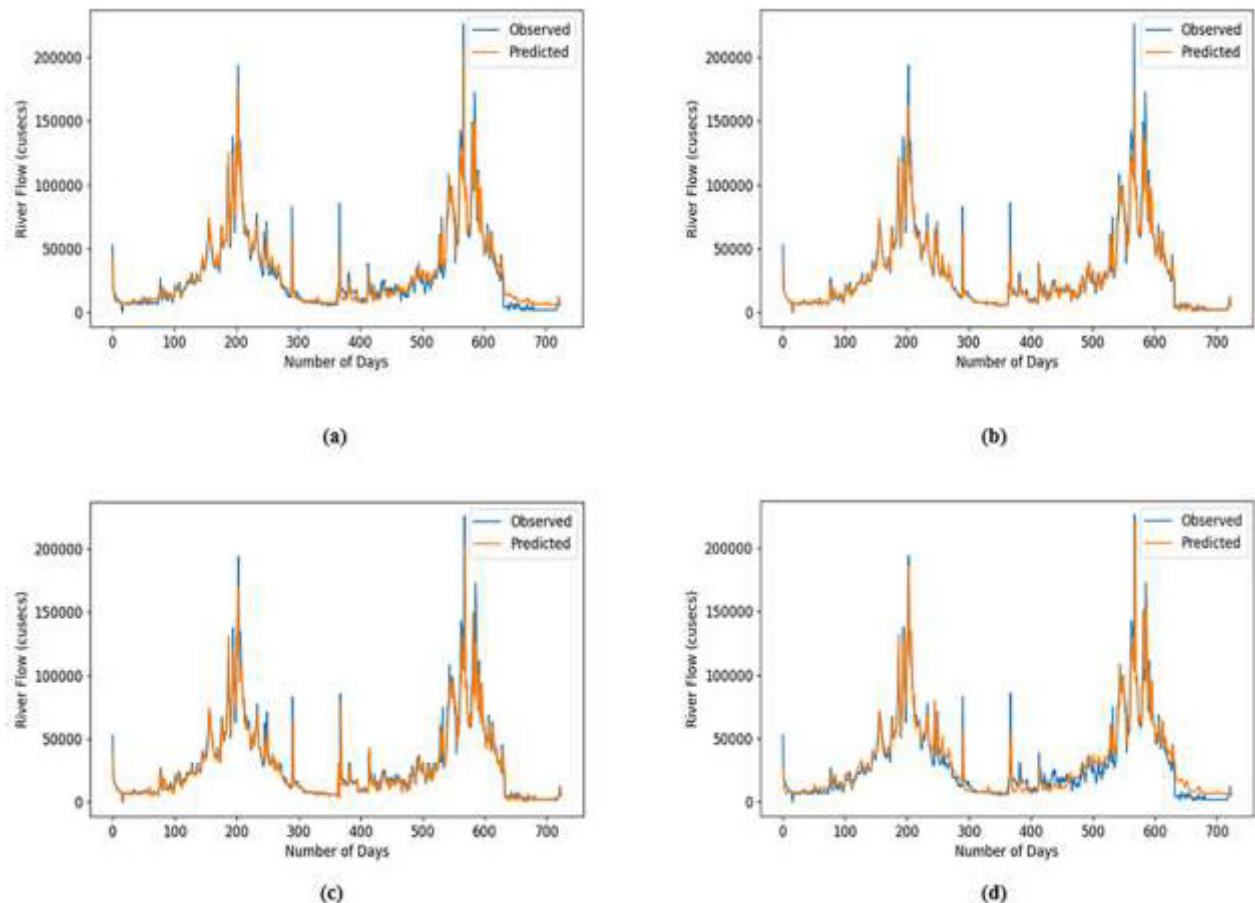


FIGURE 10. LSTM model performance for a look-back period of 4 days (a) Trial 3 with 300 epoch, 100 batch size and 0.88 R^2 (b) Trial 4 with 350 epoch, 100 batch size and 0.90 R^2 (c) Trial 5 with 300 epoch, 16 batch size and 0.91 R^2 (d) Trial 6 with 250 epoch, 128 batch size and 0.87 R^2 .

TABLE 6. Metrics performance for multiple combinations of parameters during the LSTM experiment.

Trial No.	Look-back window size	Epochs	Batch size	R^2	R	RMSE
1	2 days	300	64	0.754	0.87	15688
2	3 days	350	72	0.78	0.89	14709
3	4 days	300	100	0.88	0.94	10849
4	4 days	350	100	0.90	0.95	10032
5	4 days	300	16	0.91	0.96	9870
6	4 days	250	128	0.873	0.93	11254
7	5 days	300	128	0.78	0.89	14662
8	6 days	250	72	0.77	0.88	15218

from the LSTM experiment. Trial 6 successfully captured the highs of river flow but with a relatively low value of R^2 which is 0.87.

The impact of different hyperparameter combinations on the LSTM model performance with a look-back period of four days is illustrated in Fig. 10 a-d. Fig. 11 a-d shows a set of LSTM model train and test loss (MAE) graphs obtained during multiple trials for a look-back period of four days.

During the second experiment, the ML-GMDH model was trained for various look-back periods (from 1 to 10 days) in order to get the optimal results. A four-day look-back window was found to be the most effective. Table 7 presents the performance of the ML-GMDH model for various look-back window sizes. It can be seen in Table 7 that for look-back periods of two and three days, the R^2 value is not satisfactory which shows that the meteorological observations of the last two days cannot well describe the hydrological phenomenon on the third day. The same is true for the 5 and 6-day look-back period.

The results of trial 3 reveal that the simulated flow is in the best agreement with the observed flow on a 4-day look-back period.

Trial 3 showed the most promising results, with the highest R^2 of 0.88, and the lowest RMSE that is 10943. The

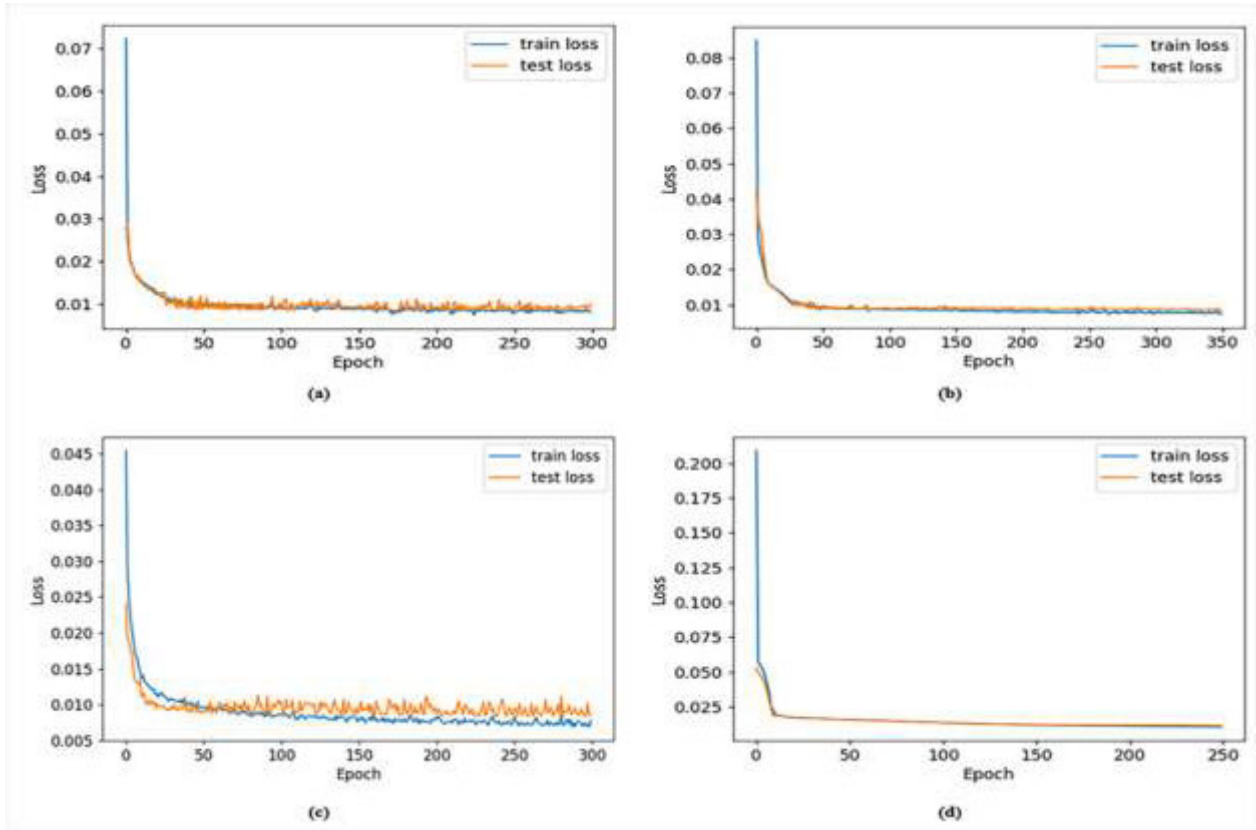


FIGURE 11. (a) Model Loss for Trial 3 with 0.88 R^2 (b) Model Loss for Trial 4 with 0.90 R^2 (c) Model Loss for Trial 5 with 0.91 R^2 (d) Model Loss for Trial 6 with 0.87 R^2 .

TABLE 7. Metrics performance for ML-GMDH experiment.

Trial No.	Look back window size	R^2	R	RMSE
1	2 days	0.74	0.88	16078
2	3 days	0.747	0.88	15904
3	4 days	0.88	0.94	10943
4	5 days	0.755	0.88	15840
5	6 days	0.75	0.88	15673

performance of the ML-GMDH model for various look-back periods during Experiment 4 is shown in Fig. 12 a-d.

Both LSTM and ML-GMDH performed best for a look-back period of four days. The LSTM model performed best, with the highest score of R^2 0.91, and minimum RMSE of 9870. The ML-GMDH model predicted the river flow with 0.88 R^2 . The ML-GMDH model performs with a somewhat lower accuracy but is still able to deliver good results. ML-GMDH has shown its capability without manual efforts

to adjust hyperparameters repeatedly to obtain the best output of the model.

The findings of the proposed hybrid methodology signify its competence and reliability in river flow forecasting. By coupling hydrological rainfall-runoff analysis with intelligent time-series modeling, this study presents an optimal set of attributes and introduces a new criterion of using catchment runoff instead of rainfall to predict downstream river flow. The existing literature does not employ the integration of runoff generated through physically based hydrological modeling with deep learning methods for river flow forecasting. This integration has led to the incorporation of the impact of static spatial characteristics of the study area into intelligent time-series modeling, and it can be observed that it has brought about significant accuracy in the results of the study. The notable observation of hydrological modeling was the high CN value that shows the Chenab catchment area has a high runoff potential and is therefore prone to flooding. This excess water can be stored in water reservoirs and utilized for hydropower generation or harvesting purposes to raise agricultural productivity [80]. LSTM is an established benchmark model for time-series forecasting [44], and ML-GMDH is also a reliable method in terms of its capabilities in time-series modeling [47], [48]. The results of the multiple statistical metrics used in this research indicate that both deep learning models are fully proficient

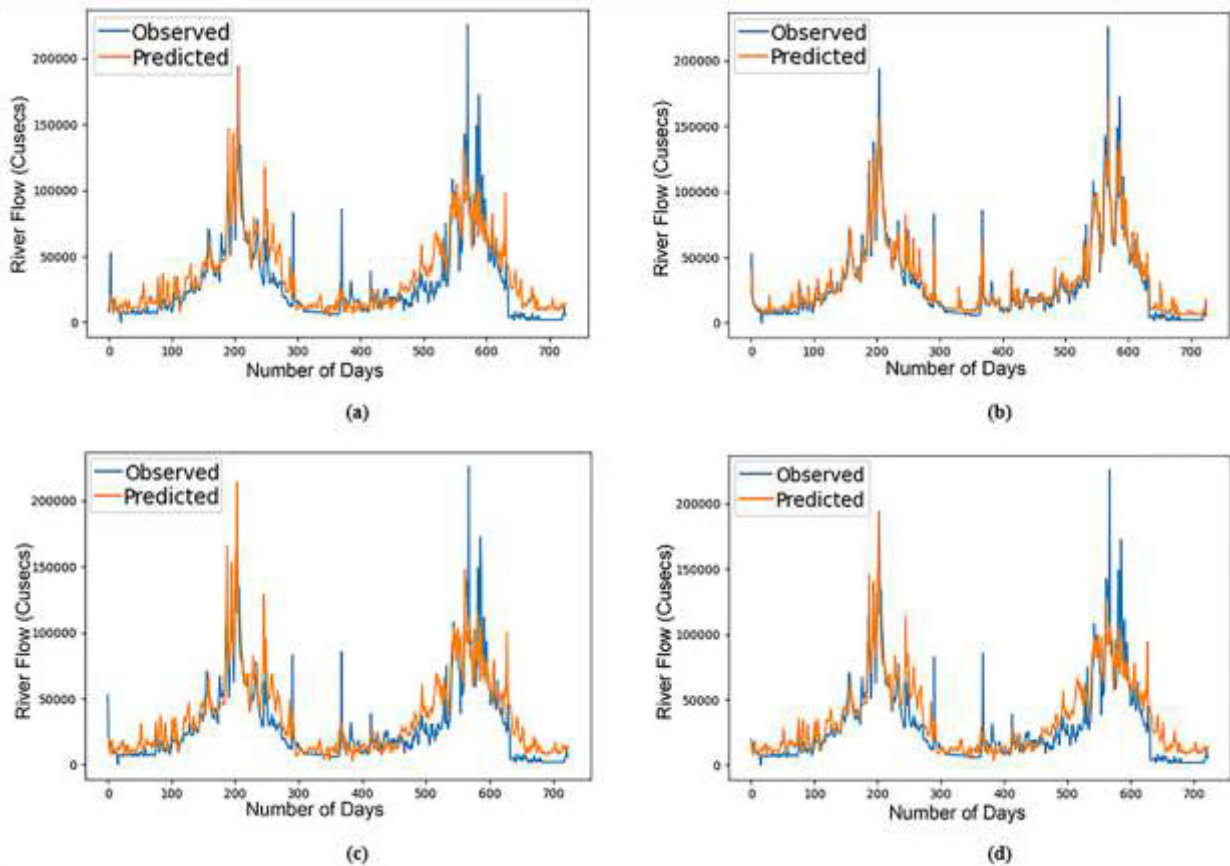


FIGURE 12. ML-GMDH model performance for different look-back periods (a) Trial 1 with 2 days look-back period and 0.74 R^2 (b) Trial 3 with 4 days look-back period and 0.88 R^2 (c) Trial 4 with 5 days look-back period and 0.75 R^2 (d) Trial 5 with 6 days look-back period and 0.755 R^2 .

in producing reliable and satisfactory results for river flow forecasting.

Currently, the Flood Forecasting Division (FFD) of Pakistan, which works under PMD, uses physically based hydrological modeling approaches and “Quantitative Precipitation Measurement (QPM)” RADAR for rain observations. The efficiency of the flood forecasting system employed by the PMD-FFD is very low. RADAR data is not calibrated and, thus, is not very reliable [30]. According to researchers and PMD documentation, systems employed by PMD-FFD need significant improvement to minimize future flood damage [30], [31]. Other studies conducted for flood forecasting purposes in the Chenab Basin, Pakistan are based on hydrological modeling [7], [37], [55]. In the absence of an advanced and effective flood forecasting system in the Chenab Basin, we hope that our contribution to flood prediction will help a number of people save lives and prevent financial losses in this area. The proposed approach can successfully predict river flow based on satellite-based observations in the absence of upstream hydrological observations. Therefore, this is a suitable solution for ungauged rivers.

One of the limitations of this study is that only 13 years data was used in this study. The system can be made more robust by adding additional years data. In the future, model can be

deployed in a production environment where it can make predictions on new, unseen data. As the ML-GMDH model does not require much manual effort and is self-optimizing in nature, we would like to test this model for flood monitoring in other basins of Pakistan and the world. The proposed approach can be further improved by using hourly datasets for flood monitoring. Increased precision of output could result in more time to make informed decisions. In the future, more computationally efficient AI methods, such as the “Gated Recurrent Unit (GRU)” can be used to predict river flow.

V. CONCLUSION

This study proposes an effective hybrid method that couples hydrological modeling with intelligent time-series analysis to foresee upcoming floods. The integration of hydrological and deep learning modeling has yielded significant results. The flood risk possibilities are assessed using some influential parameters. Meteorological and physical variables, such as temperature, humidity, wind speed, soil, and land use, are connected to natural processes such as evaporation, infiltration, and transpiration, which in turn impact the process of flooding. The spatial characteristics of an area directly affect the hydrological processes. The proposed coupled approach

generates an optimal set of attributes to predict river flow. The final set of attributes used in this study to forecast daily river flow were runoff, temperature, atmospheric pressure, and humidity.

The proposed approach is applied specifically to predict the daily river flow at the Marala Station, situated along the Chenab River in Pakistan which is the busiest and most heavily populated area of Pakistan and has been facing frequent flooding for the last two decades. An advanced and effective flood forecasting system was an essential requirement of the Chenab basin. The proposed hybrid approach demonstrated promising results and proved to be a reliable solution.

The lack of forecasting and communication of early warning information is always one of the major reasons for extensive disasters. Timely forecasts of floods can save lives, livestock, and property as they help in adopting precautionary measures in advance. We hope that our contribution to flood prediction will help many people save lives and avoid structural damage.

- The results indicated that both deep learning models are capable of producing reliable and satisfactory results for river flow prediction. The LSTM model outperformed the ML-GMDH model with 0.91 R^2 and the lowest RMSE. ML-GMDH signifies its forecasting ability with 0.88 R^2 .
- The proposed methodology is a suitable choice for ungauged river basins, aims to predict river flow based on meteorological and physical information in the absence of upstream hydrologic data.
- Deep learning-based flood forecasting solutions that employ freely available satellite data are a cost-effective solution to the problem with a minimum requirement of investment and resources.
- An effective flood forecasting system was an essential requirement of the Chenab basin, where due to the absence of machine learning and deep learning flood forecasting models, the advanced solution was an urgent need.
- In the future, the proposed models can be deployed in a production environment where they can make predictions on new, unseen data, thereby helping save lives and prevent financial losses in this area.

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